College of Engineering Trivandrum

Department of Computer Applications

20MCA203 . Design and Analysis of Algorithms (Semester 3)

Tutorial II

1. Solve the recurrence relation, $T(n)=2T(n/2) + bn \lg n$ by substitution method.
2. Solve the recurrence relation, $T(n)=3T(n/4) + \Theta(n^2)$ by recursion tree method
3. Consider implementing binary search on:

   (a) A sorted array of n elements (b) A sorted singly linked list of n elements

   For each implementation:

   (i) Write the recurrence relation for the running time.

   (ii) Solve the recurrence and determine the asymptotic time complexity.

   (iii) Explain why binary search has different time complexities for an array & a linked list